

What You'll Learn

This workbook builds computational thinking skills step by step. Each activity teaches ONE new concept, then combines it with what came before — so by Activity 11, you're using everything together, and by Activity 12, YOU are the designer.

THE 12 ACTIVITIES

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| 1 What is an Algorithm?
Learn what makes a set of instructions an algorithm. | 2 Sequencing Puzzle
Discover why the ORDER of steps matters. |
| 3 Spot the Loop
Recognize repetition in songs, chores, and patterns. | 4 Write Your Own Loop
Use "REPEAT N TIMES" to shorten programs. |
| 5 Yes/No Conditionals
Make programs decide with IF / THEN / OTHERWISE. | 6 Nested Conditionals
Chain multiple Yes/No questions into decision trees. |
| 7 Debug the Robot
Find bugs — sequence, missing-step, and wrong-value. | 8 Bubble Sort
Learn your first NAMED algorithm. |
| 9 Selection Sort
Discover that the same problem has multiple solutions. | 10 Pattern Recognition
Find the RULE that predicts the future. |
| 11 Treasure Hunt Grid
Combine everything to navigate a puzzle. | 12 Design Your Own Algorithm
YOU'RE the designer. Build from scratch. |

SKILLS YOU'LL BUILD

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| ● Computational thinking — breaking problems into parts and building solutions | ● Logical reasoning — following rules to reach conclusions |
| ● Problem decomposition — turning big challenges into smaller steps | ● Pattern recognition — spotting rules and predicting outcomes |
| ● Debugging skills — finding and fixing mistakes | ● Creative problem-solving — designing your own solutions |

Ready? Turn the page and let's begin!